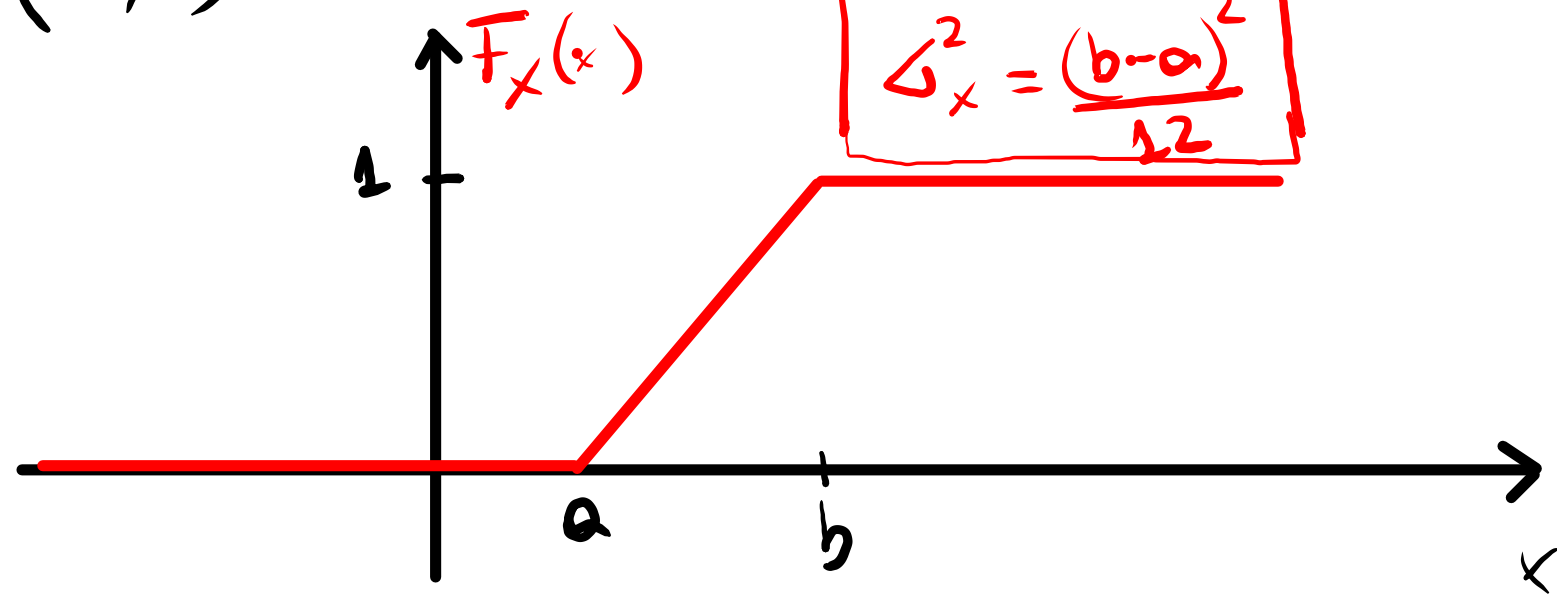
UNIFORME: $X \sim Unif(a,b)$ con $a < b$

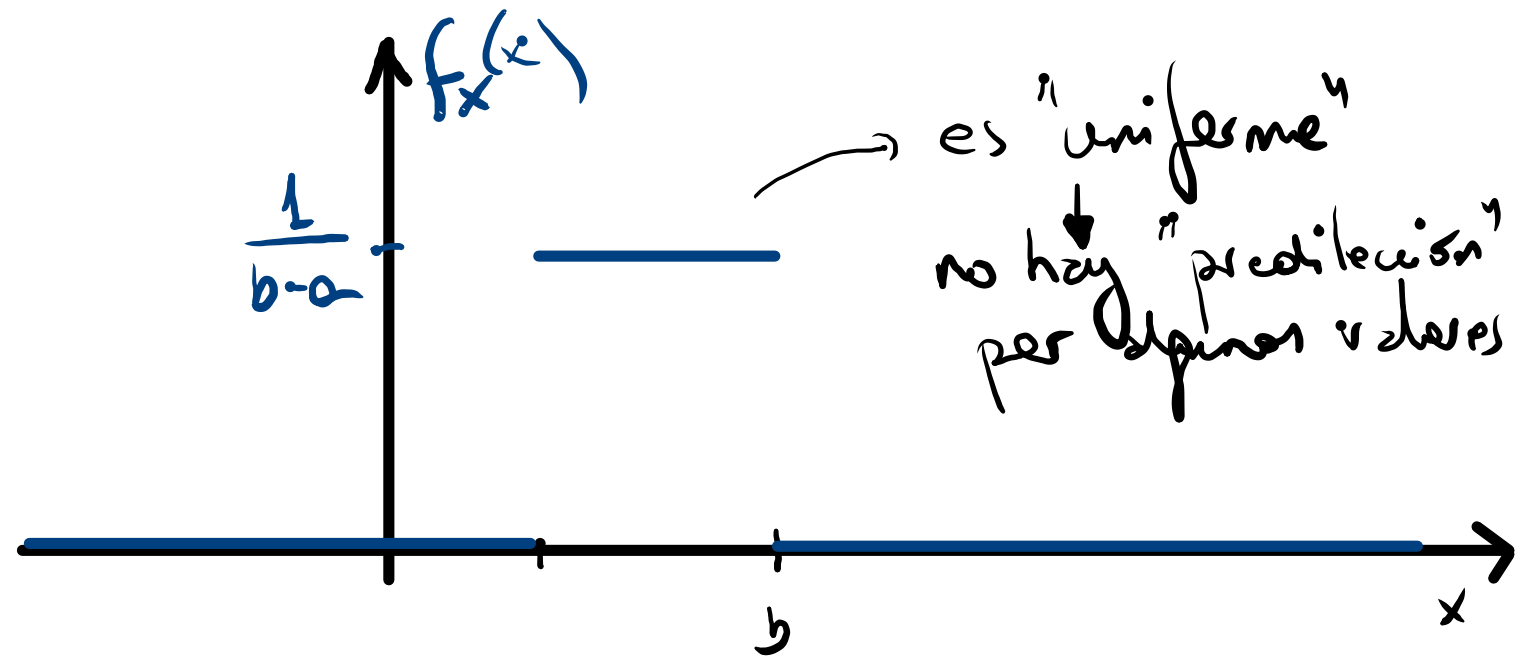
$$\mu_X = \frac{a+b}{2}$$

$$\sigma_X^2 = \frac{(b-a)^2}{12}$$

$$F_X(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \geq b \end{cases}$$



$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in (a,b) \\ 0 & x \notin (a,b) \end{cases}$$



es "uniforme"

no hay "predilección" por algunos valores

$$\mu_X = E[X] = \int_{-\infty}^{+\infty} x\, f_X(x)\, dx = \underbrace{\cancel{\int_{-\infty}^{a}}}_{0} + \int_a^b + \underbrace{\cancel{\int_b^{+\infty}}}_{0} = \int_a^b x \cdot \frac{1}{b-a}\, dx =$$

$$= \frac{x^2}{2(b-a)}\Big|_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{b+a}{2} = \frac{a+b}{2} \rightarrow \boxed{\mu_X = \frac{a+b}{2}}$$

$$E[X^2] = \int_a^b x^2 \frac{1}{b-a}\, dx = \frac{x^3}{3(b-a)}\Big|_a^b = \frac{b^3-a^3}{3(b-a)} = \frac{b^2+ba+a^2}{3}$$

$$\sigma_X^2 = Var(X) = E[X^2] - \mu_X^2 = \frac{b^2+ba+a^2}{3} - \left(\frac{b^2+2ab+a^2}{4}\right) = \frac{b^2-2ba+a^2}{12} = \frac{(b-a)^2}{12}$$

$$\boxed{\sigma_X^2 = \frac{(b-a)^2}{12}}$$